

Department of Computer Science & Engineering

I Year CSE II Sem

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Computer Networks and The Internet

Contents : Introduction to Computer Networks, Advantages of computer networks

- Types of Networks : Local Area Network, Metropolitan Area Network, Campus/Corporate Area Network, Personal Area Network, Peer-to-Peer Networks
- Physical Components Of network
- Network Metrics: Downloading, Uploading, Buffering, Speed-Baud rate, Bandwidth
- Wired media: Twisted Pair wires, Coaxial cables, Fiber Optic Cables, 5G-Technologies
- Wireless Media: Terrestrial Microwaves, Satellite Communication, Infrared Communication
- Networking Devices : HUB, Repeater, Switch, Bridge , Router, Gateway, Network Interface card, RJ45
- Network Topologies: Bus Topology, Ring Topology, Mesh Toplogy, Hybrid Topology

Introduction to Computer Networks

Objective:

To be acquainted with:

- The definitions of networking
- Network topology
- Network peripherals, hardware and software

Definitions

1.1 Network Definition

- A network can be defined as two or more computers connected together in such a way that they can share resources.
- The purpose of a network is to share resources.

A resource may be:

- A file
- A folder
- A printer
- A disk drive
- Or just about anything else that exists on a computer.

- A network is simply a collection of computers or other hardware devices that are connected together, either physically or logically, using special hardware and software,
- To allow them to exchange information and cooperate. Networking is the term that describes the processes involved in designing, implementing, upgrading, managing and otherwise working with networks and network technologies.

Advantages of Networking

- Connectivity and Communication
- Data Sharing
- Hardware Sharing
- Internet Access
- Internet Access Sharing
- Data Security and Management
- Performance Enhancement and Balancing
- Entertainment

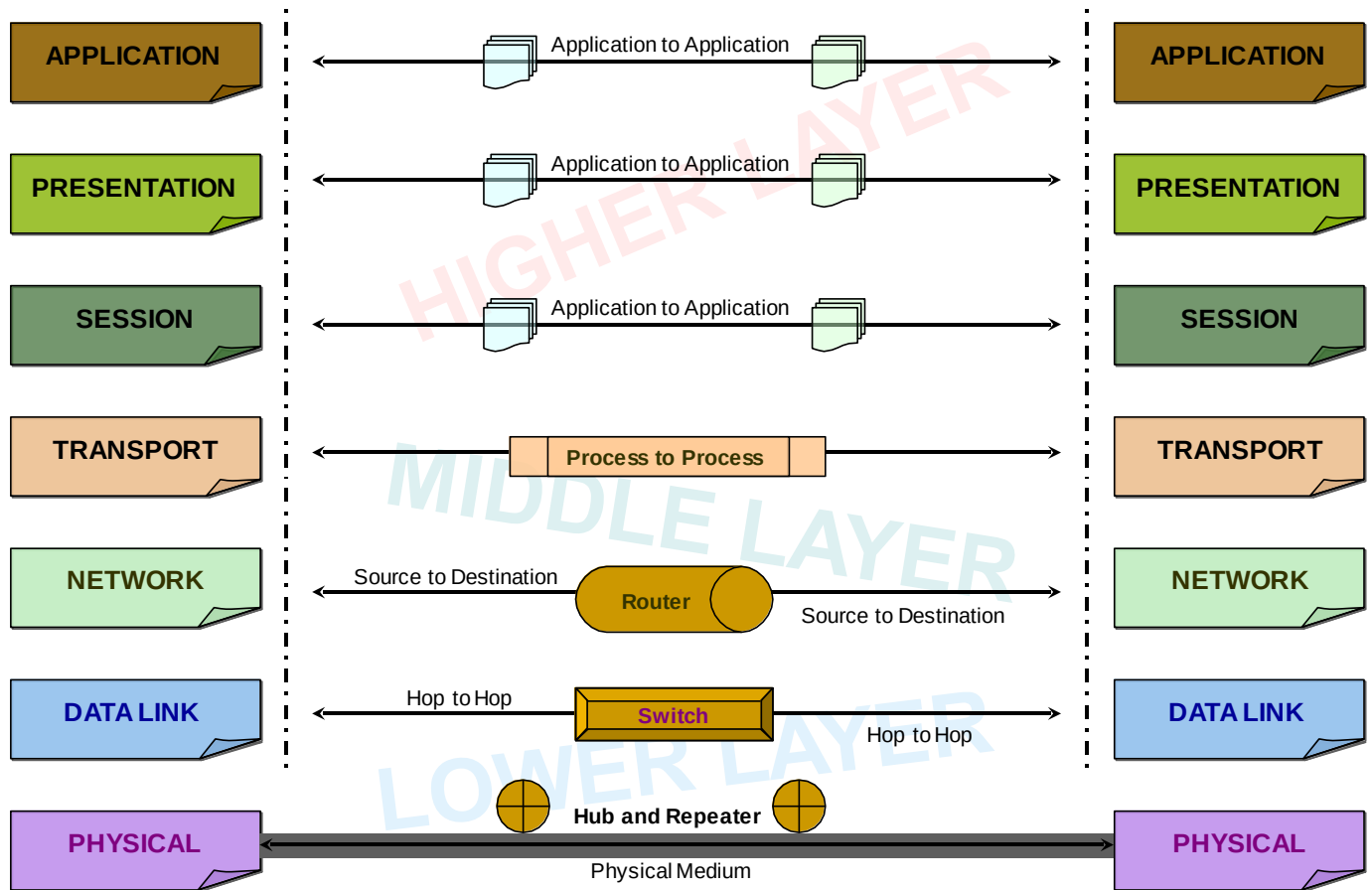
The Disadvantages (Costs) of Networking

- Network Hardware, Software and Setup Costs
- Hardware and Software Management and Administration Costs
- Undesirable Sharing
- Illegal or Undesirable Behavior
- Data Security Concerns

Open Systems Interconnection Reference Model(ISO-OSI)

- Open Systems Interconnection Basic Reference Model is an abstract description for layered communications and computer network protocol design. It was developed as part of the **Open Systems Interconnection (OSI)** initiative. In its most basic form, it divides network architecture into seven layers which, from top to bottom, are the Application, Presentation, Session, Transport, Network, Data-Link, and Physical Layers. It is therefore often referred to as the **OSI Seven Layer Model**.

OSI Model's 7 Layers

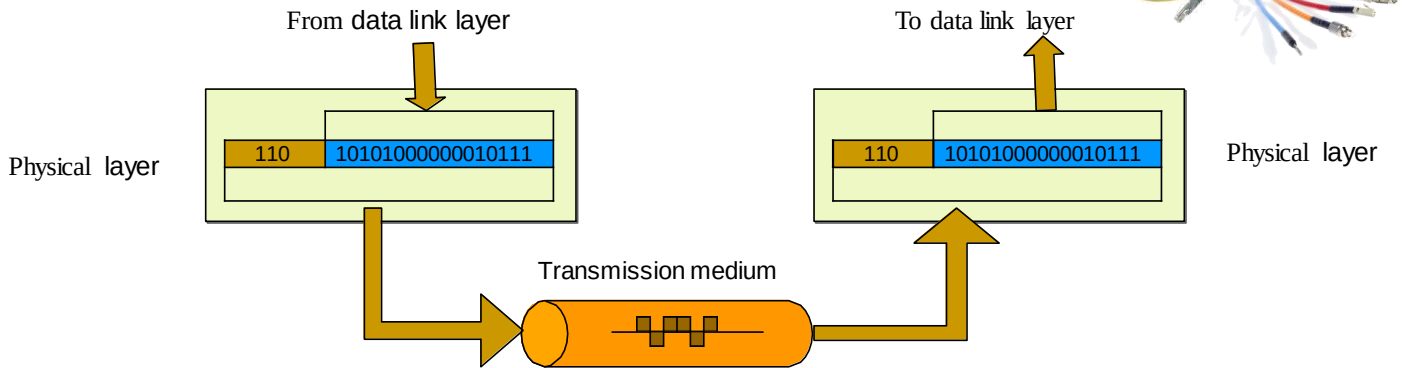


Data, Protocol & Activities

OSI Layers	TCP/IP Suit	Activities
Application	Application Telnet, FTP, SMTP, HTTP, DNS, SNMP, <i>Specific address</i> etc...	To allow access to network resources
Presentation	Presentation	To Translate, encrypt, and compress data
Session	Session	To establish, manage, and terminate session
Transport	Transport SCTP, TCP, UDP, Sockets and <i>Ports address</i>	To Provide reliable process-to-process Message delivery and error recovery
Network	Network IP, ARP/RARP, ICMP, IGMP, <i>Logical address</i>	To move packets from source to destination; to provide internetworking
Data Link	Data Link IEEE 802 Standards, TR, FDDI, PPP, <i>Physical address</i>	To organize bits into frames; to provide Hop-to-hop delivery
Physical	Physical Medium, Coax, Fiber, 10base, Wireless	To Transmit bits over a medium; to provide Mechanical and electrical specifications

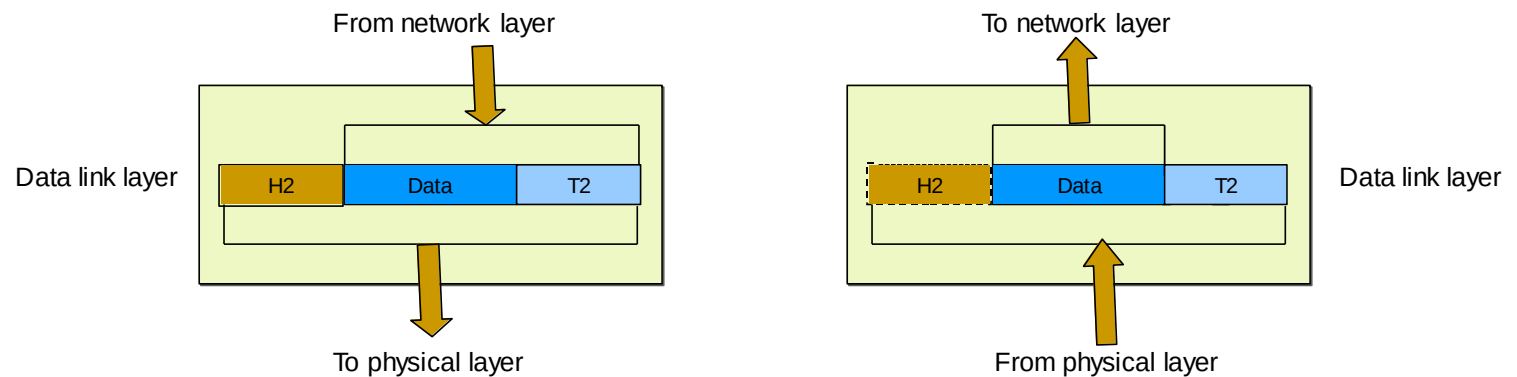
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Physical Layer



- One of the major function of the physical layer is to move data in the form of electromagnetic signals across a transmission medium.
- Its responsible for movements of individual bits from one hop (Node) to next. Both data and the signals can be either *analog* or *digital*.
- Transmission media work by conducting energy along a physical path which can be wired or wireless
- Concerned:
 - Physical characteristics of interface and medium (Transmission medium)
 - Representation of bits (stream of bits (0s or 1s) with no interpretation and encoded into signals) Data rate (duration of a bit, which is how long it last)
 - Synchronization of bits (sender and receivers clock must be synchronized) Line configuration (Point-to-Point, Point-to-Multipoint)
 - Physical topology
 - Transmission mode (Simplex, half duplex, full duplex)

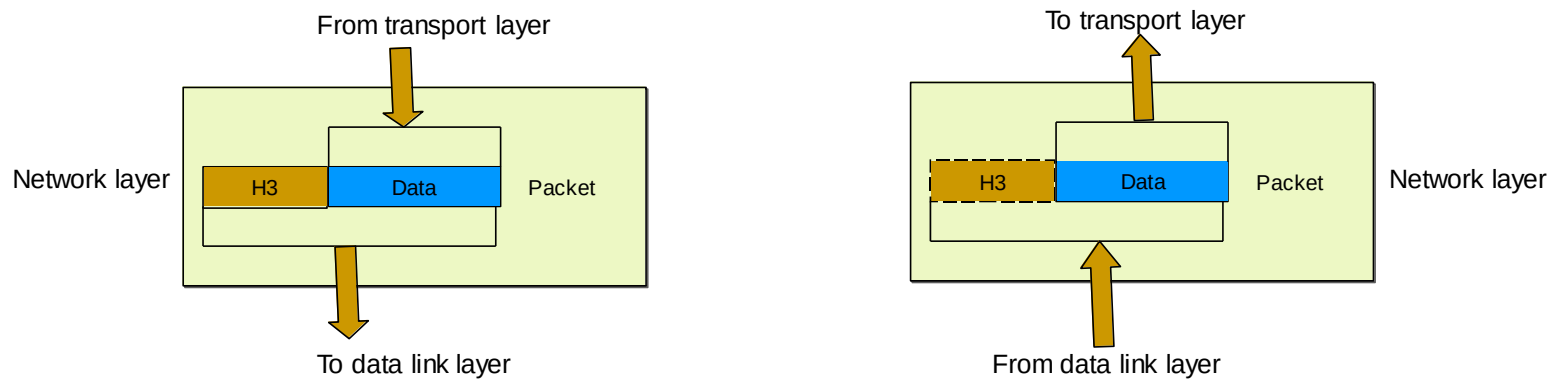
Data Link Layer (Host to Host)



- Data link layer is responsible for moving frames from one hop (Node) to the next.
- Concerned:
 - Framing (stream of bits into manageable data units)
 - Physical addressing (MAC Address)
 - Flow Control (mechanism for overwhelming the receiver)
 - Error Control (trailer, retransmission)
 - Access Control (defining master device in the same link)

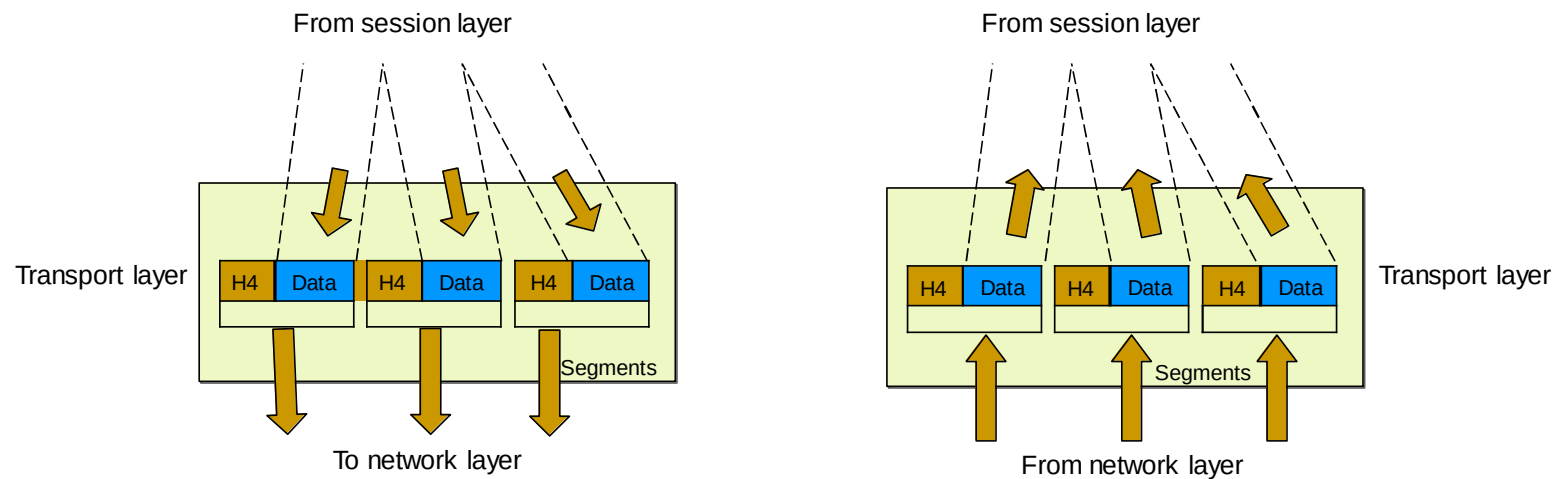
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Network Layer (Source to Destination)



- The network layer is responsible for the delivery of individual packets from the source host to the destination host.
- Concerned:
 - Logical addressing (IP Address)
 - Routing (Source to destination transmission between networks)

Transport Layer (Process to Process)



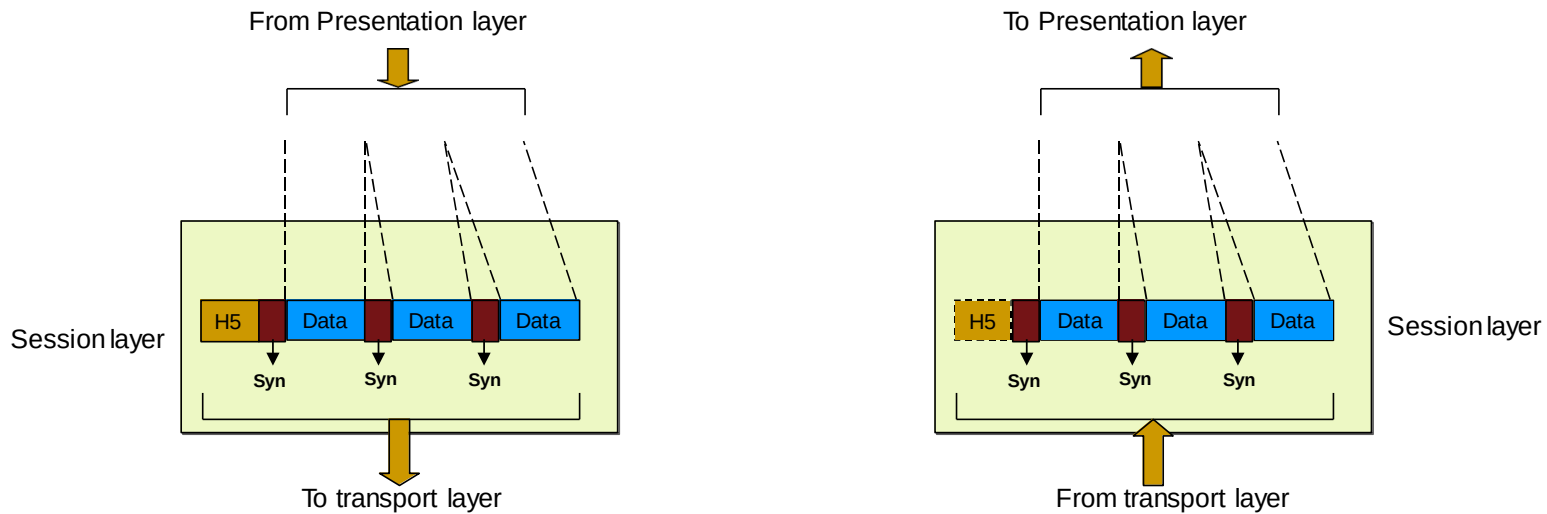
- The transport layer is responsible for the delivery of a message from one process to another

Concerned:

- •Service-point addressing (Port address)
- .Segmentation and reassembly (Sequence number)
- Connection control (Connectionless or connection oriented) Flow control (end to end)
- Error Control (Process to Process)

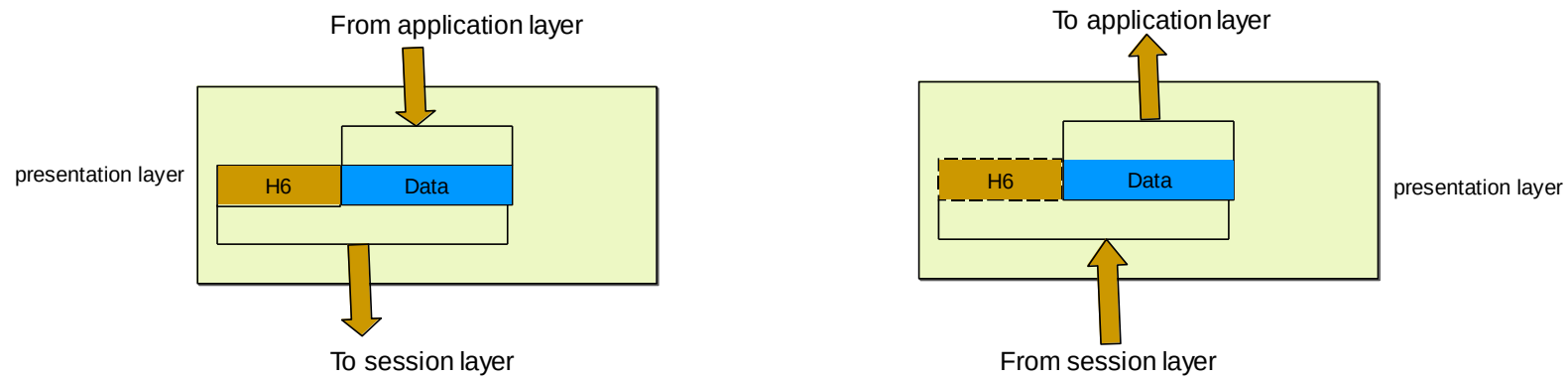
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Session Layer (Dialog initiation)



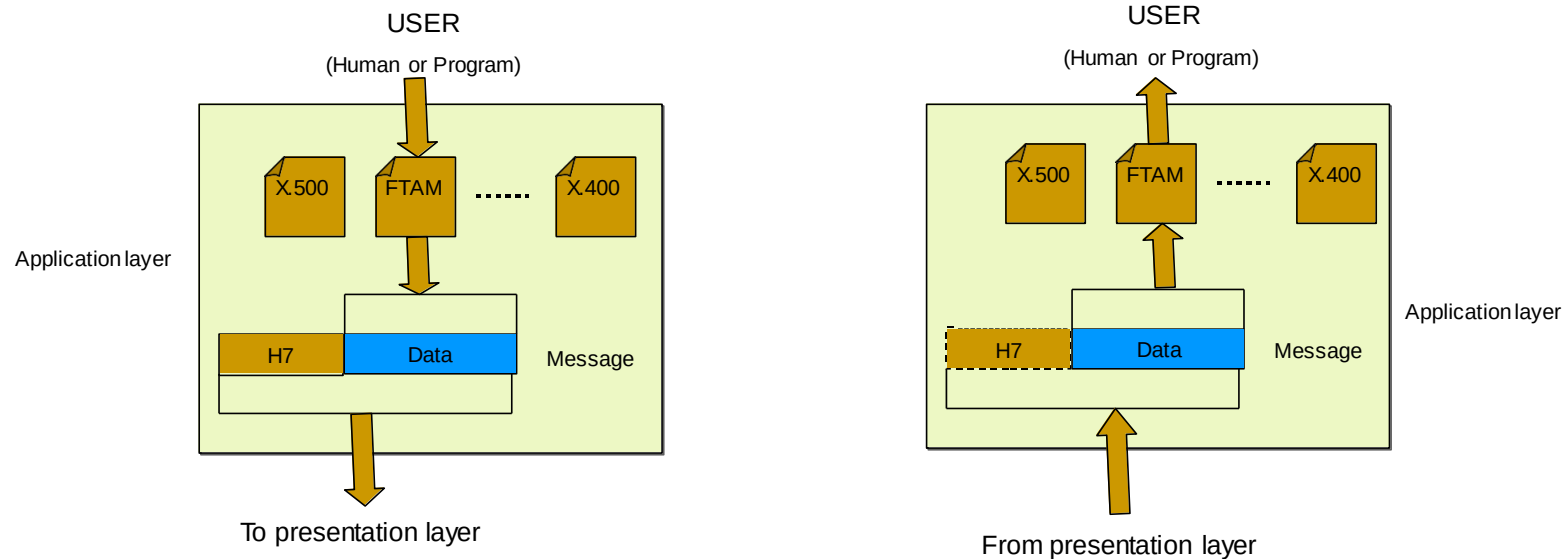
- The session layer is responsible for dialog control and synchronization
- **Concerned:**
 - •Dialog Control (Half Duplex/Full duplex)
 - Synchronization (Synchronization points, process inline within same page)

Presentation Layer (dependency)



- The presentation layer is responsible for translation, compression and encryption
- Concerned:
 - Translation (interoperability between different encoding system)
 - Encryption (Privacy schemes)
 - Compression (data compression)

Application Layer (user level service)



- The application layer is responsible for providing services to the user.
- Concerned:
 - Network virtual terminal (Software) File transfer, access and management Mail services
 - Directory services (access to distributed database sources for global information about various objects and services)
 -

Network Devices

- Switch

- Split large networks into small segments, decreasing the number of users sharing the same network resources and bandwidth.
- Understands when two devices want to talk to each other, and gives them a *switched* connection
- Helps prevent data collisions and reduces network congestion, increasing network performance.
- Most home users get very little, if any, advantage from switches, even when sharing a broadband connection.
- Resides on Layer 2 of the OSI model.

OSI Model Layers

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical



Contd...

- **Bridge**

- Connects two LANs and forwards or filters data packets between them.
- Creates an extended network in which any two workstations on the linked LANs can share data.
- Transparent to protocols and to higher level devices like routers.
- Forward data depending on the Hardware (MAC) address, not the Network address (IP).
- Resides on Layer 2 of the OSI model.

OSI Model Layers

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical



Contd...

- Repeater

- Used to boost the signal between two cable segments or wireless access points.
- Can not connect different network architecture.
- Does not simply amplify the signal, it regenerates the packets and retimes them.
- Resides on Layer 1 of the OSI model.

OSI Model Layers

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical



Contd...

- Router

- A device that connects any number of LANs.
- Uses standardized protocols to move packets efficiently to their destination.
- More sophisticated than bridges, connecting networks of different types (for example, star and token ring)
- Forwards data depending on the Network address (IP), not the Hardware (MAC) address.
- Routers are the only one of these four devices that will allow you to share a single IP address among multiple network clients.
- Resides on Layer 3 of the OSI model.

OSI Model Layers

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical



Contd..

- Additional Network Hardware Devices

- Network Interface Cards (NICs)
 - Puts the data into packets and transmits packet onto the network.
 - May be wired or wireless.
- Gateway
 - Connects networks with different protocols like TCP/IP network and IPX/SPX networks.
 - Routers and Gateways often refer to the same device.
- Proxy server
 - Isolates internal network computers from the internet.
 - The user first access the proxy server and the proxy server accesses the internet and retrieves the requested web page or document. The user then gets a copy of that page from the proxy server.



HUB

- A network hub is a node that broadcasts data to every computer or Ethernet-based device connected to it. A hub is less sophisticated than a switch, the latter of which can isolate data transmissions to specific devices.
- Network hubs are best suited for small, simple local area network (LAN) environments.

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Hubs cannot provide routing capabilities or other advanced network services. Because they operate by forwarding packets across all ports indiscriminately, network hubs are sometimes referred to as "dumb switches."



RJ45

- A registered jack (RJ) is a standardized physical network interface for connecting telecommunications or data equipment. The physical connectors that registered jacks use are mainly of the modular connector and 50-pin miniature ribbon connector types. The most common twisted-pair connector is an 8-position, 8-contact (8P8C) modular plug and jack commonly referred to as an RJ45 connector.

Cont....

RJ45 is a type of connector commonly used for Ethernet networking. It looks similar to a telephone jack, but is slightly wider. Since Ethernet cables have an RJ45 connector on each end, Ethernet cables are sometimes also called RJ45 cables.

The "RJ" in RJ45 stands for "registered jack," since it is a standardized networking interface. The "45" simply refers to the number of the interface standard. Each RJ45 connector has eight pins, which means an RJ45 cable contains eight separate wires. If you look closely at the end of an Ethernet cable, you can actually see the eight wires, which are each a different color. Four of them are solid colors, while the other four are striped.



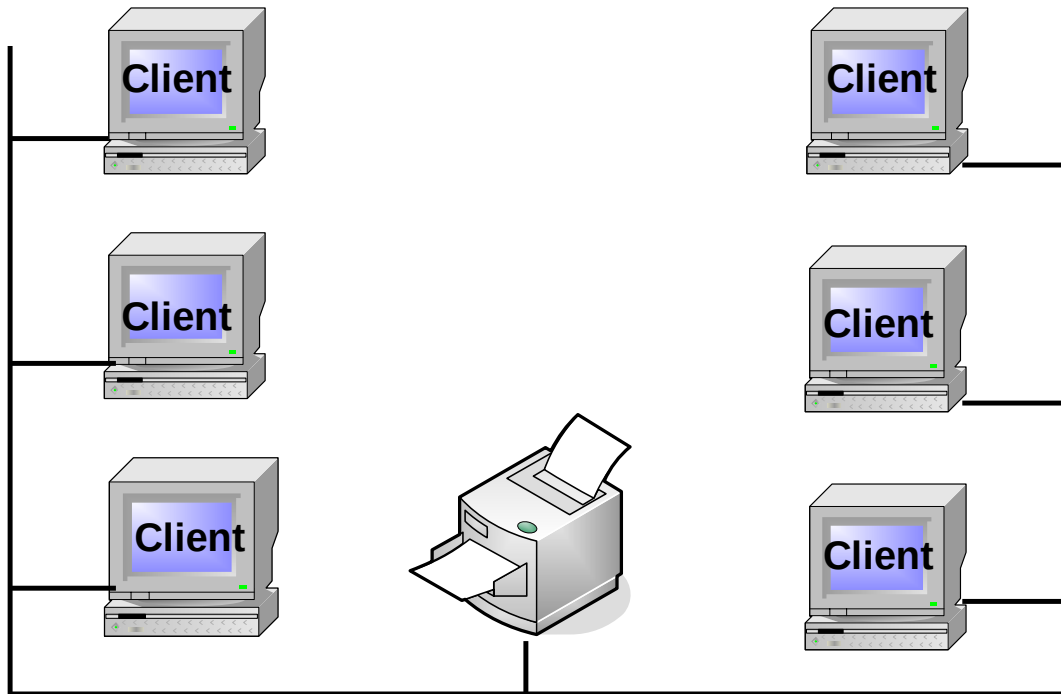
Types of networks

- 1.LAN (Local Area Network)**
- 2.MAN (Metropolitan Area Network)**
- 3.WAN(Wide Area Network)**
- 4.CAN(Campus Area Network)**
- 5.PAN(Personal Area Network)**
- 6.Peer to peer Networks**

Local Area Networks (LANs):

- **A local area network (LAN)** is a computer network covering a small geographic area, like a home, office, or group of buildings

The Local Network (LAN)



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LAN

- Most LANs connect to the Internet at a central point: a router. Home LANs often use a single router, while LANs in larger spaces may additionally use network switches for more efficient packet delivery.
- LANs almost always use Ethernet, WiFi, or both in order to connect devices within the network. Ethernet is a protocol for physical network connections that requires the use of Ethernet cables. WiFi is a protocol for connecting to a network via radio waves.
- A variety of devices can connect to LANs, including servers, desktop computers, laptops, printers, IoT devices, and even game consoles. In offices, LANs are often used to provide shared access to internal employees to connected printers or servers.

Metropolitan Area Network (MAN)

- A metropolitan area network (MAN) is a network that interconnects users with computer resources in a geographic area or region larger than that covered by even a large local area network (LAN) but smaller than the area covered by a wide area network (WAN). The term is applied to the interconnection of networks in a city into a single larger network (which may then also offer efficient connection to a wide area network). It is also used to mean the interconnection of several local area networks by bridging them with backbone lines. The latter usage is also sometimes referred to as a campus network

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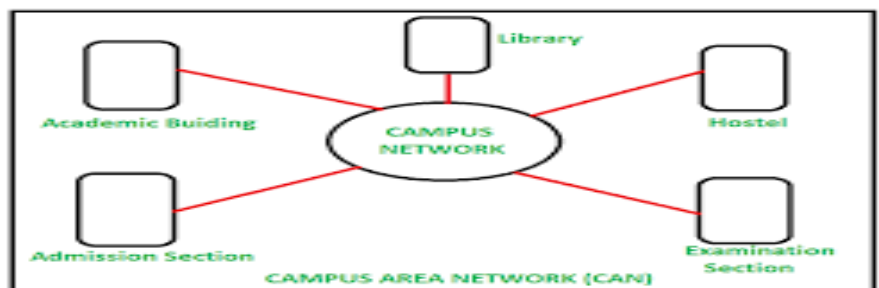
Personal Area Network

A personal area network (PAN) connects electronic devices within a user's immediate area. The size of a PAN ranges from a few centimeters to a few meters. One of the most common real-world examples of a PAN is the connection between a Bluetooth earpiece and a smart phone. PANs can also connect laptops, tablets, printers, keyboards, and other computerized devices.



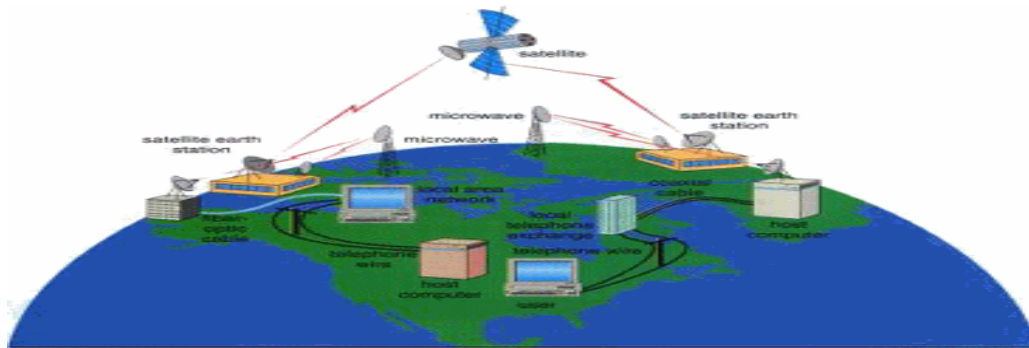
Campus Area Network

- A campus area network (CAN) is a computer network that spans a limited geographic area. CANs interconnect multiple local area networks (LAN) within an educational or corporate campus. Most CANs connect to the public Internet



Wide Area Network

- **Wide Area Network (WAN)** is a computer network that covers a broad area (i.e., any network whose communications links cross metropolitan, regional, or national boundaries). Or, less formally, a network that uses routers and public communications links
- The largest and most well-known example of a WAN is the Internet.
- WANs are used to connect LANs and other types of networks together, so that users and computers in one location can communicate with users and computers in other locations



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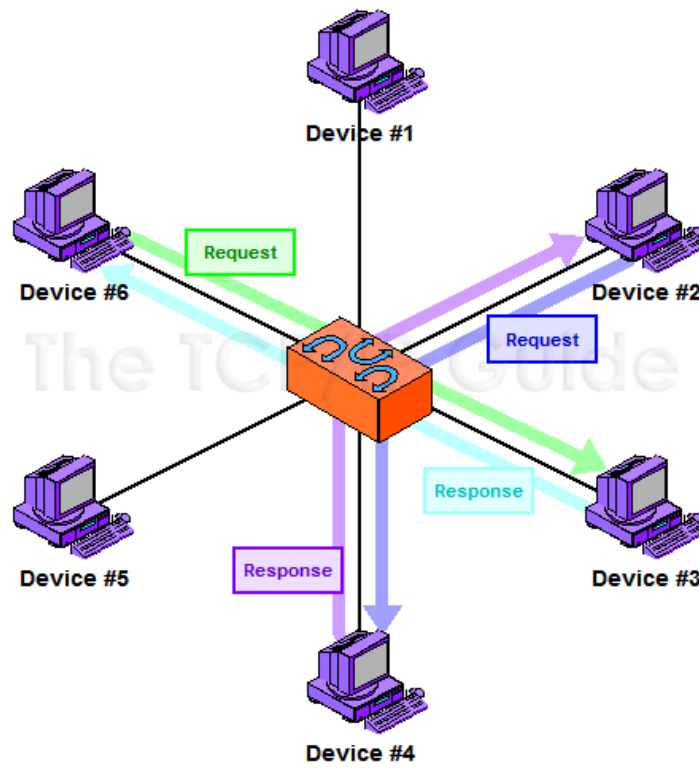
Intranet and Internet Specifications

- **Intranet:** An intranet is a private network that is contained within an enterprise. It may consist of many interlinked local area networks and also use leased lines in the wide area network.
- An intranet uses TCP/IP, HTTP, and other Internet protocols and in general looks like a private version of the Internet. With tunneling, companies can send private messages through the public network, using the public network with special encryption/decryption and other security safeguards to connect one part of their intranet to another.
- **Internet:** is a worldwide system of computer networks - a network of networks in which users at any one computer can, if they have permission, get information from any other computer (and sometimes talk directly to users at other computers).

Peer-to peer network

- A peer-to-peer network is a network where the computers act as both workstations and servers.
- great for small, simple, and inexpensive networks.
- In a strict peer-to-peer networking setup, every computer is an equal, a *peer* in the network.
- Each machine can have resources that are shared with any other machine.
- There is no assigned role for any particular device, and each of the devices usually runs similar software. Any device can and will send requests to any other.

Peer-to peer network



Network topology

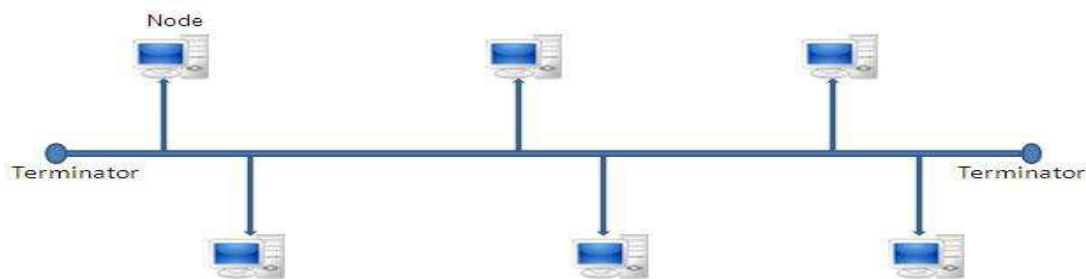
- A topology is a way of “laying out” the network. Topologies can be either physical or logical.
- Physical topologies describe how the cables are run.
- Logical topologies describe how the network messages travel

Network topology

- Bus (can be both logical and physical)
- Star (physical only)
- Ring (can be both logical and physical)
- Mesh (can be both logical and physical)
- Hybrid

Bus Topology

- A bus is the simplest physical topology. It consists of a single cable that runs to every workstation
- This topology uses the least amount of cabling, but also covers the shortest amount of distance.



Bus topology

- it is difficult to add a workstation
- have to completely reroute the cable and possibly run two additional lengths of it.
- if any one of the cables breaks, the entire network is disrupted. Therefore, it is very expensive to maintain.

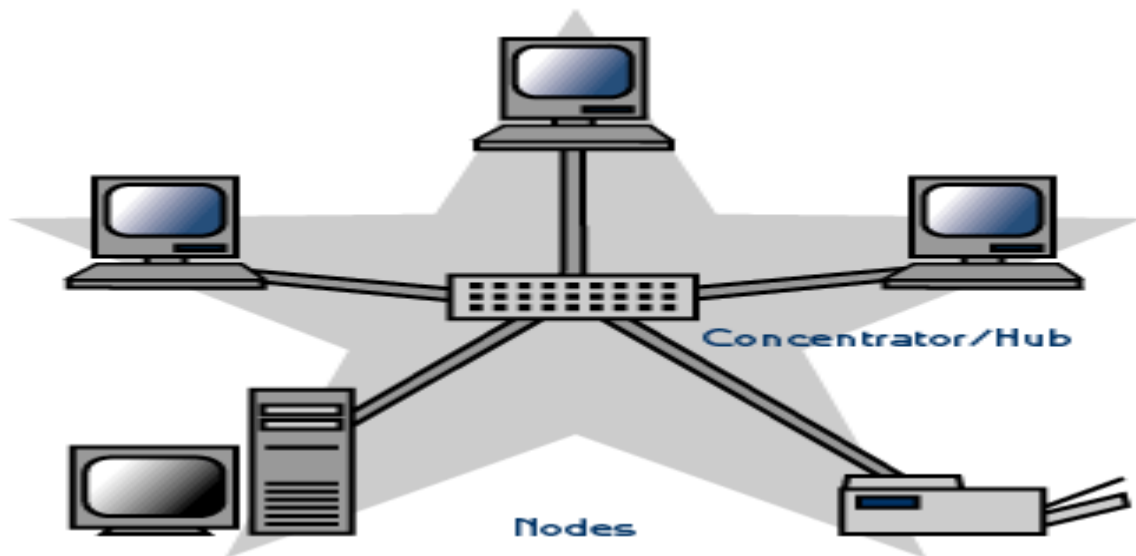
Star Topology

- A physical star topology branches each network device off a central device called a hub, making it very easy to add a new workstation.
- Also, if any workstation goes down it does not affect the entire network. (But, as you might expect, if the central device goes down, the entire network goes down.)
- Some types of Ethernet and ARCNet use a physical star topology.

Star Topology

- Star topologies are easy to install. A cable is run from each workstation to the hub. The hub is placed in a central location in the office.
- Star topologies are more expensive to install than bus networks, because there are several more cables that need to be installed, plus the cost of the hubs that are needed.

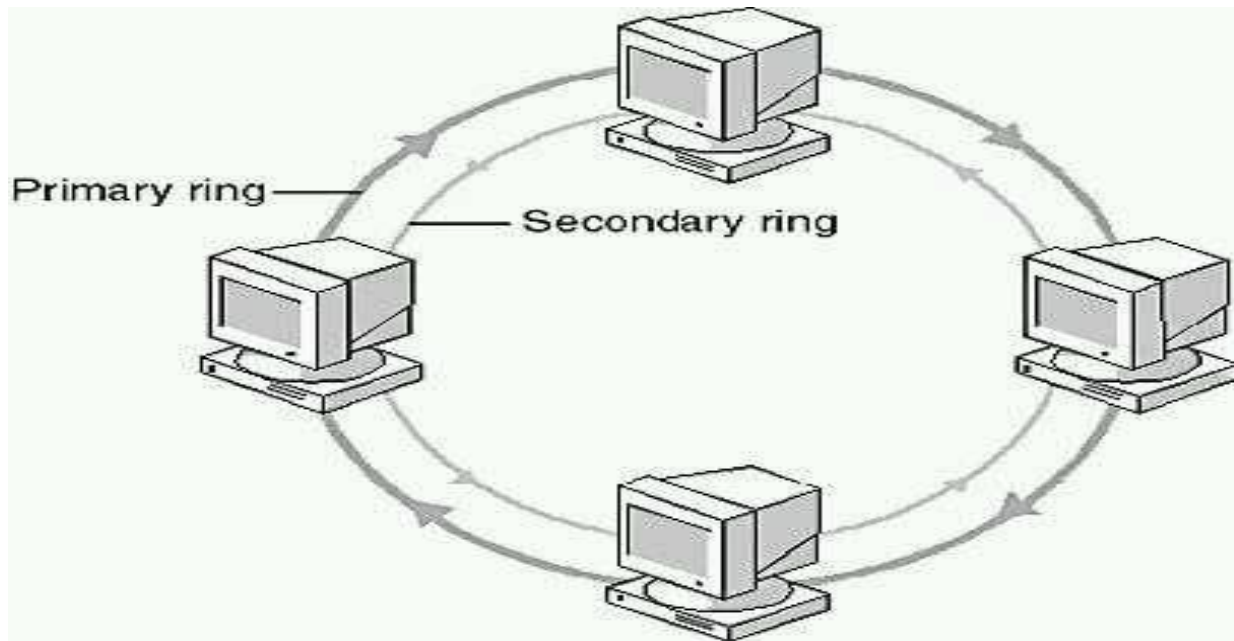
Star Topology



Ring Topology

- Each computer connects to two other computers, joining them in a circle creating a unidirectional path where messages move workstation to workstation.
- Each entity participating in the ring reads a message, then regenerates it and hands it to its neighbor on a different network cable.

Ring Topology



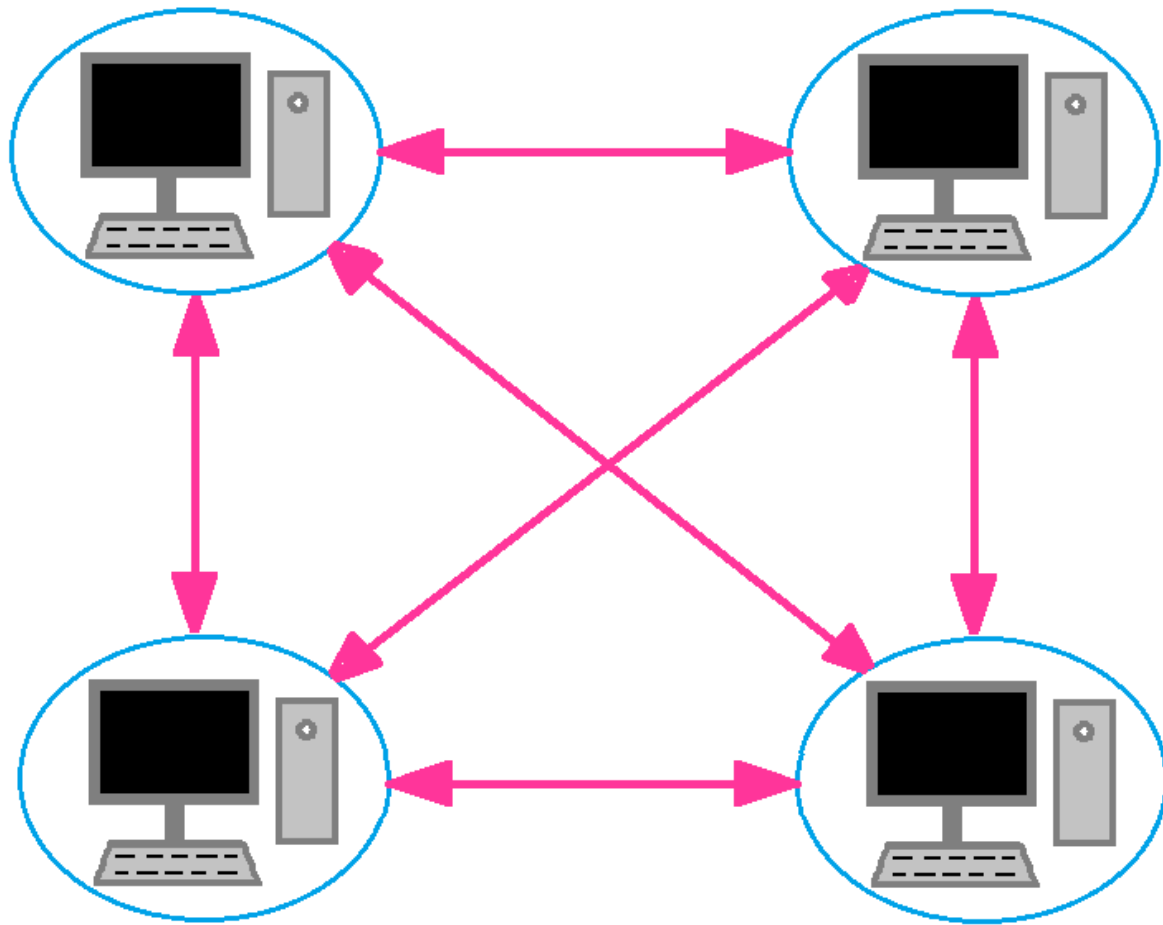
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Ring Topology

- The ring makes it difficult to add new computers.
- Unlike a star topology network, the ring topology network will go down if one entity is removed from the ring.
- Physical ring topology systems don't exist much anymore, mainly because the hardware involved was fairly expensive and the fault tolerance was very low.

Mesh Topology

- The mesh topology is the simplest logical topology in terms of data flow, but it is the most complex in terms of physical design.
- In this physical topology, each device is connected to every other device
- This topology is rarely found in LANs, mainly because of the complexity of the cabling.
- If there are x computers, there will be $(x \times (x-1)) \div 2$ cables in the network. For example, if you have five computers in a mesh network, it will use $5 \times (5 - 1) \div 2$, which equals 10 cables. This complexity is compounded when you add another workstation.

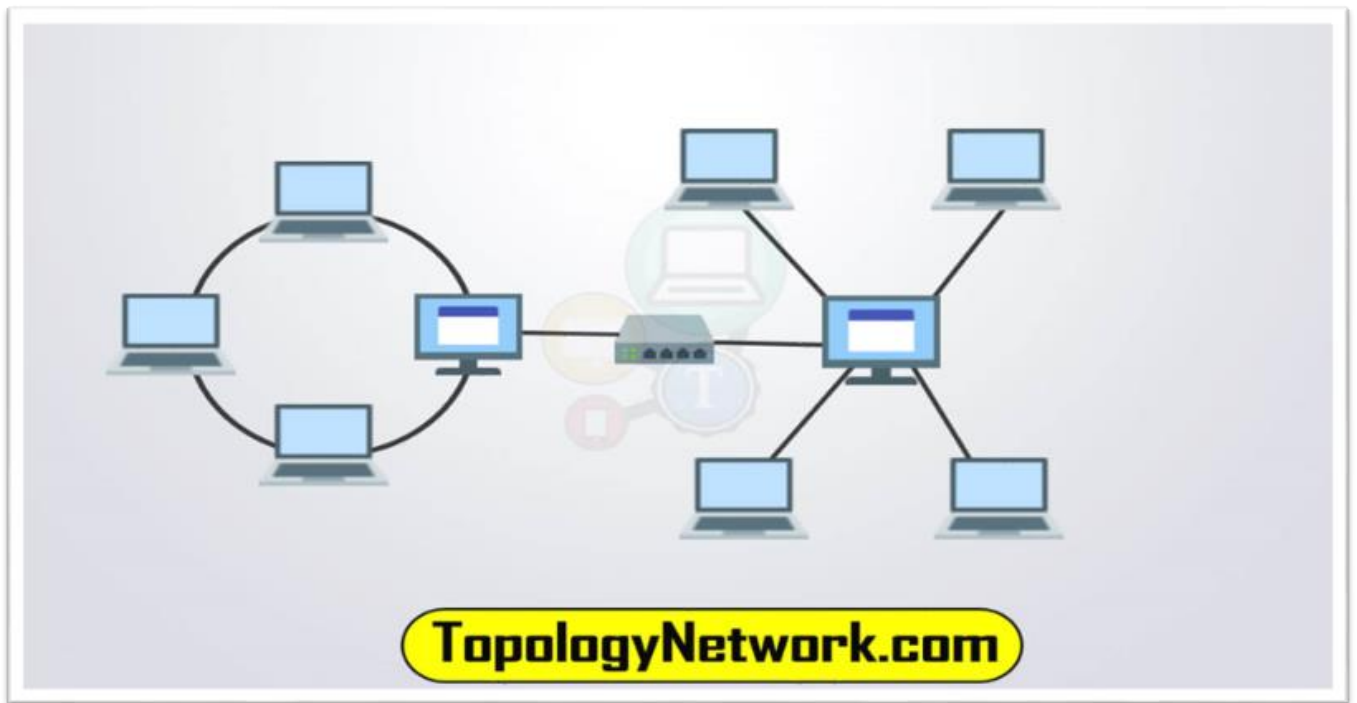


Mesh Topology

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Hybrid topology:

A hybrid topology is a type of network topology that uses two or more differing network topologies. These topologies can include a mix of bus topology, mesh topology, ring topology, star topology, and tree topology



Network Metrics

Network metrics are qualitative and quantitative ways to observe and determine network behavior.

Network performance monitoring tools are designed to deliver out-of-the-box visibility into network availability and performance metrics. They're also designed to be fast, lightweight, and easy to use. To monitor network metrics, you can set alerts and get real-time notifications for outages and potential errors

This way, you can make more informed decisions to minimize downtime and fix network failures. In addition, some of these tools allow you to monitor and track network bandwidth, network configurations,

Contd...

- VoIP devices, and IP addresses. These comprehensive monitoring tools include advanced capabilities designed to ensure your network is functioning correctly and delivering uninterrupted services. The features of these tools can include the following:
 - Built-in reporting system
 - Bandwidth utilization monitoring
 - Intuitive centralized dashboards

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- Network traffic monitoring
- Flow-based monitoring and reporting
- Dynamic and intelligent network mapping
- Automatic alerting
- User experience monitoring
- Automatic recovery
- Remote administration
- Wireless network performance metrics monitoring

Downloading

- Downloading is the transmission of a file or data from one computer to another over a network, usually from a larger server to a user device. Download can refer to the general transfer of data or to transferring a specific file. It can also be called to download, DL or D/L.

Uploading

- Uploading is the process of putting web pages, images and files onto a web server.
- Uploading (U/L) refers to the process of copying files from a smaller peripheral device to a large central system. This process may involved transferring data from a local computer to a remote computer (and usually large) system, or transferring data from a computer to a bulletin board system (BBS).

Difference between Downloading and Uploading

- Uploading means data is being sent from your computer to the Internet.
- Examples of uploading include sending email, posting photos on a social media site and using your webcam. Even clicking on a link on a web page sends a tiny data upload.
- Downloading means your computer is receiving data from the Internet.

Cont....

- Examples of downloading include opening a web page, receiving email, purchasing music files and watching online videos.

Buffering

- Buffering is the practice of pre-loading segments of data when streaming video content. Streaming — the continuous transmission of audio or video files from a server to a client — is the process that makes watching videos online possible. Buffering helps make streaming run more smoothly because videos can start playing before the entire video is loaded.

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- Buffering is a core component of the streaming process. However, many viewers do not realize this and are only aware of the buffering process when it happens slowly or interrupts streaming. This is why viewers may use the word "buffering" to describe delays in content loading.

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- Buffering can be compared to a grocery store's inventory and shelving process. Stores maintain inventory so they can easily restock shelves and avoid inconveniencing customers. Grocery store customers generally go about their shopping without thinking about the stocking process unless an item they want is unavailable. In the same way, video players pre-load video segments to avoid disrupting the viewing experience.

Speed Baud Rate

- BaudRate: Specify the rate at which bits are transmitted
- You configure BaudRate as bits per second. The transferred bits include the start bit, the data bits, the parity bit (if used), and the stop bits. However, only the data bits are stored.

Cont...

- The baud rate is the rate at which information is transferred in a communication channel. In the serial port context, "9600 baud" means that the serial port is capable of transferring a maximum of 9600 bits per second. If the information unit is one baud (one bit), then the bit rate and the baud rate are identical. If one baud is given as 10 bits, (for example, eight data bits plus two framing bits)

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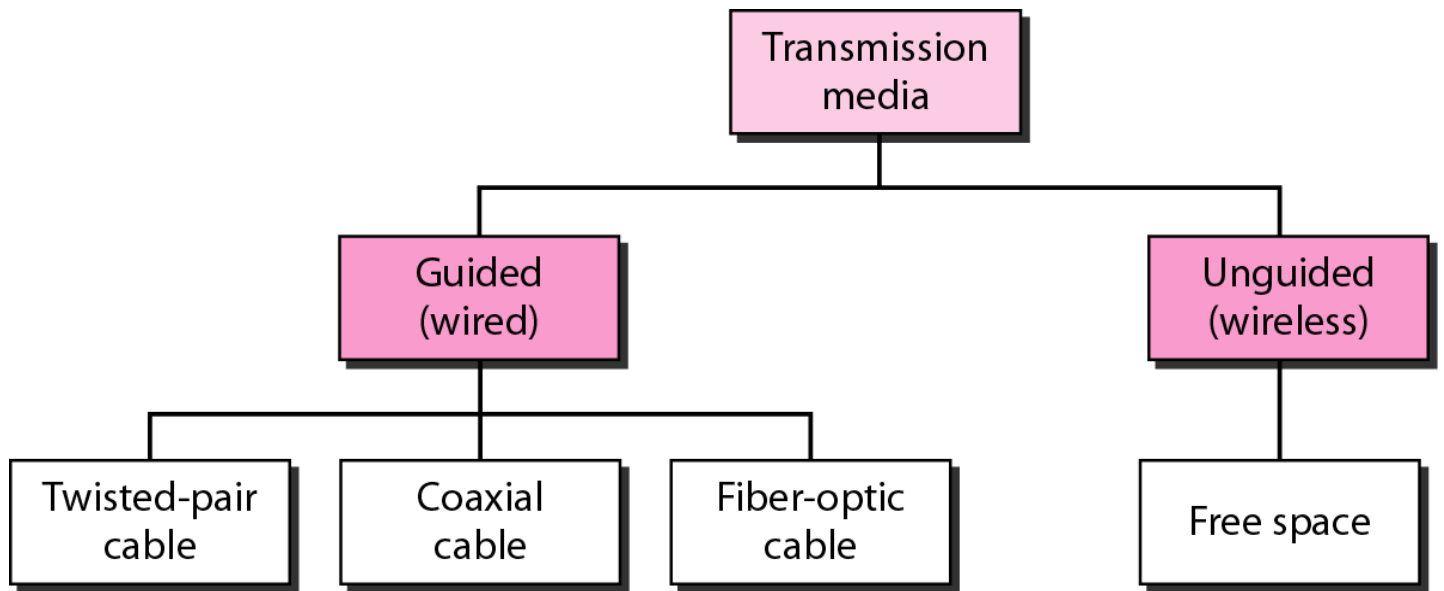
- the bit rate is still 9600 but the baud rate is $9600/10$, or 960. You always configure Baud Rate as bits per second. Therefore, in the above example, set Baud Rate to 9600.

Bandwidth

- Network bandwidth is a measure of the data transfer rate or capacity of a given network. It's a crucial network measurement for understanding the speed and quality of a network.

Network bandwidth is commonly measured in bits per second (bps). In practice, organizations and internet service providers (ISPs) measure bandwidth in megabits per second (Mbps) or gigabits per second (Gbps).

Classes of transmission media



- Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.

Twisted pair cable

- Consists of two insulated copper wires arranged in a regular spiral pattern to minimize the electromagnetic interference between adjacent pairs
- Often used at customer facilities and also over distances to carry voice as well as data communications
- Low frequency transmission medium

Twisted pair cable



Types of twisted-pair cable

- STP (shielded twisted pair)
 - the pair is wrapped with metallic foil or braid to insulate the pair from electromagnetic interference
- UTP (unshielded twisted pair)
 - each wire is insulated with plastic wrap, but the pair is encased in an outer covering

Twisted Pair Advantages

- Inexpensive and readily available
- Flexible and light weight
- Easy to work with and install

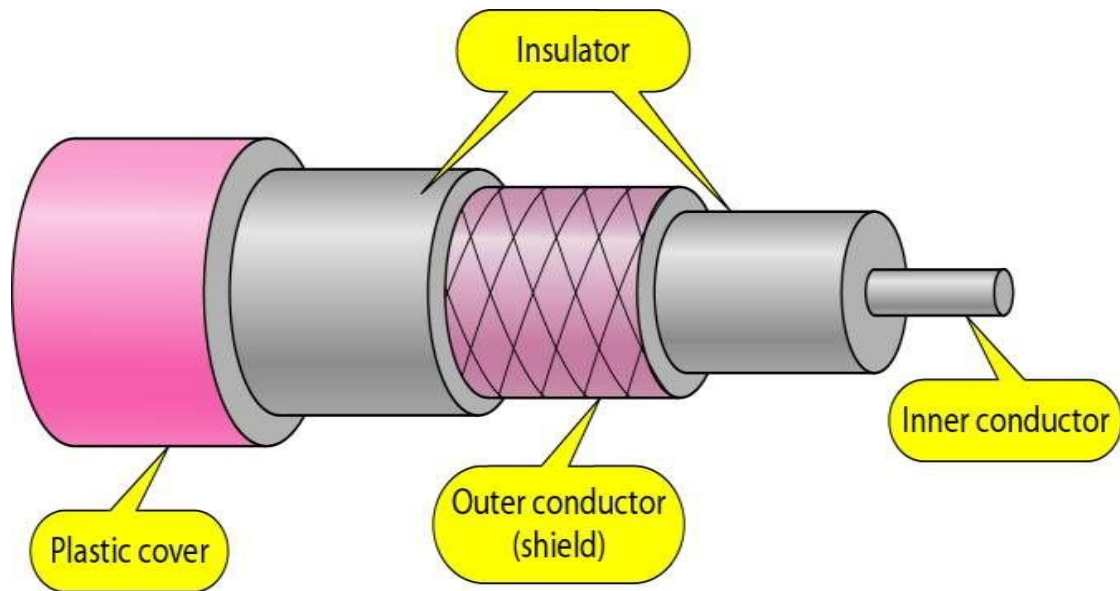
Twisted Pair Disadvantages

- Susceptibility to interference and noise
- Attenuation problem
 - For analog, repeaters needed every 5-6km
 - For digital, repeaters needed every 2-3km
- Relatively low bandwidth (3000Hz)

Coaxial Cable (or Coax)

- Used for cable television, LANs, telephony
- Has an inner conductor surrounded by a braided mesh
- Both conductors share a common center axial, hence the term “co-axial”

Coaxial Cable (or Coax)



Coax Advantages

- Higher bandwidth
 - 400 to 600Mhz
 - up to 10,800 voice conversations
- Can be tapped easily (pros and cons)
- Much less susceptible to interference than twisted pair

Coax Disadvantages

- High attenuation rate makes it expensive over long distance
- Bulky

Fiber Optic Cable

- Relatively new transmission medium used by telephone companies in place of long-distance trunk lines
- Also used by private companies in implementing local data communications networks
- Require a light source with injection laser diode (ILD) or light-emitting diodes (LED)

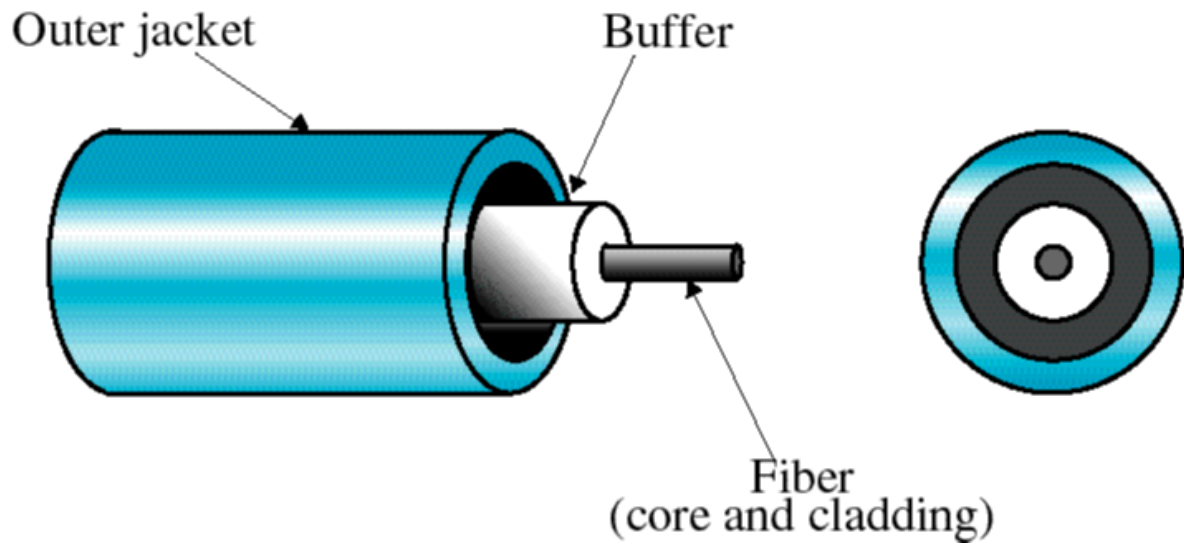
Fiber Optic Types

- multimode step-index fiber
 - the reflective walls of the fiber move the light pulses to the receiver
- multimode graded-index fiber
 - acts to refract the light toward the center of the fiber by variations in the density
- single mode fiber
 - the light is guided down the center of an extremely narrow core

Fiber Optic Advantages

- greater capacity (bandwidth of up to 2 Gbps)
- smaller size and lighter weight
- lower attenuation
- immunity to environmental interference
- highly secure due to tap difficulty and lack of signal radiation

Fiber Optic Cable



Fiber Optic Disadvantages

- expensive over short distance
- requires highly skilled installers
- adding additional nodes is difficult

Wireless (Unguided Media) Transmission

- transmission and reception are achieved by means of an antenna
- directional
 - transmitting antenna puts out focused beam
 - transmitter and receiver must be aligned
- omnidirectional
 - signal spreads out in all directions
 - can be received by many antennas

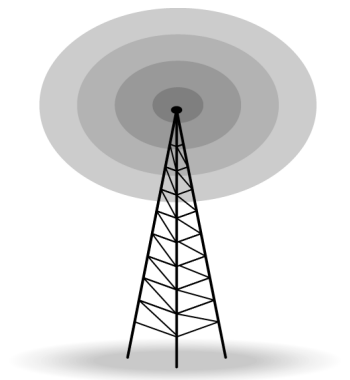
RADIO WAVES:

Radio waves are used for multicast communications, such as radio and television.

They can penetrate through walls.

Highly regulated. Use Omni directional antennas

Frequencies between 3 kHz and 1Ghz



Radio

- radio is omni-directional and microwave is directional
- Radio is a general term often used to encompass frequencies in the range 3 kHz to 300 GHz.
- Mobile telephony occupies several frequency bands just under 1 GHz.

MICROWAVES:

Microwaves are used for unicast communication such as cellular telephones,
satellite networks, and wireless LANs.

Higher frequency ranges cannot penetrate walls.

Use directional antennas - point to point line of sight communications.

1 and 300 GHz

Infrared

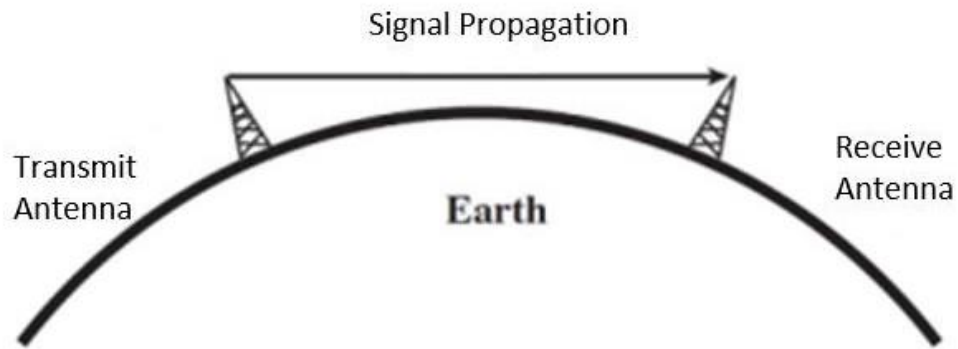
- Uses transmitters/receivers (transceivers) that modulate noncoherent infrared light.
- Transceivers must be within line of sight of each other (directly or via reflection).
- Unlike microwaves, infrared does not penetrate walls

300 GHz to 400 THz

- Terrestrial Microwave
- It is a technology which transmits the focused beam of a radio signal from one ground-based microwave transmission antenna to another antenna.
- Microwaves are generally an electromagnetic wave which has the frequency in the range from 1GHz to 1000 GHz.

- These are unidirectional waves, whereas the sending and receiving antenna is to be aligned which means the antennas are narrowly focused.
- Here antennas are mounted on the towers to send a beam to another antenna which is present at km away.
- It works on the line-of-sight transmission, which means the antennas mounted on the towers are at the direct sight of each other.

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Satellite Communication

- In general terms, a **satellite** is a smaller object that revolves around a larger object in space. For example, moon is a natural satellite of earth.
- We know that **Communication** refers to the exchange (sharing) of information between two or more entities, through any medium or channel. In other words, it is nothing but sending, receiving and processing of information.

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